

Union

Programming for Problem Solving (PPS)

GTU # 3110003

USING

{C}

Programming



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What is Union?

- ▶ Union is a **user defined data type** similar like Structure.
- ▶ It holds different data types in the **same memory location**.
- ▶ You can define a **union** with various members, but only one member can hold a value at any given time.
- ▶ Union provide an efficient way of using the same memory location for multiple-purpose.

Syntax to Define and Access Union

- ▶ Declaration of union must start with the keyword **union** followed by the union name and union's member variables are declared within braces.

Syntax

```
1 union union_name
2 {
3     member1_declaration;
4     member2_declaration;
5     . . .
6     memberN_declaration;
7 };
```

union_name is name of custom type.

memberN_declaration is individual member declaration.

- ▶ Accessing the union members:

- ➔ You need to create an object of union to access its members.
- ➔ Object is a variable of type union. Union members are accessed using the **dot operator(.)** between union's object and union's member name.

Syntax

```
1 union union_name union_variable;
```

Example to Define Union

Example

```
1 union student
2 {
3     char name[30]; // Student Name
4     int roll_no; // Student Roll No
5     float CPI; // Student CPI
6     int backlog; // Student Backlog
7 } student1;
```

- ▶ You must terminate union definition with **semicolon ;**.
- ▶ You cannot assign value to members inside the union definition, it will cause compilation error.

Example

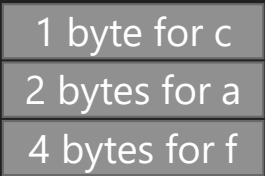
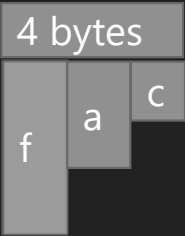
```
1 union student
2 {
3     char name[30] = "ABC"; // Student Name
4     . . .
5 } student1;
```

Structure Vs. Union

COMPARISON

STRUCTURE

UNION

Basic	The separate memory location is allotted to each member of the structure.	All members of the 'union' share the same memory location.
keyword	'struct'	'union'
Size	Size of Structure = sum of size of all the data members.	Size of Union = size of the largest member.
Store Value	Stores distinct values for all the members.	Stores same value for all the members.
At a Time	A structure stores multiple values, of the different members, of the structure.	A union stores a single value at a time for all members.
Declaration	<pre>struct ss { int a; float f; char c };</pre> 	<pre>union uu { int a; float f; char c };</pre> 

Where Union should be used?

- ▶ Mouse Programming
- ▶ Embedded Programming
- ▶ Low Level System Programming



Thank you

